

$$\mathbf{5 \quad B} \quad M_b u_b + 0 = M_b v_b + M_p v_p$$

$$(5)u_b + 0 = (5)v_b + (1)v_p \quad (1)$$

$$u_b - 0 = v_p - v_b$$

$$u_b = 3.33 - v_b \quad (2)$$

Using (1) and (2) to solve for $u_b = 1.998 \text{ m s}^{-1}$

6 B Take moments about the pivot